



CONTINGENCY LESSONS

Mathematics for SylvanSync™

Elementary



GRADES 3-5

Mathematics for SylvanSync™
Grades 3–5
VersaTiles: Add and Subtract With Regrouping

Objectives

The student will be able to:

- find the sum of two, three, or four 2- to 3-digit numbers with regrouping.
- find the difference of two 2- to 4-digit numbers with regrouping.

Materials

- *VersaTiles: Add and Subtract With Regrouping* book (VT3ASR)
- VersaTiles

Background Information

VersaTiles let you learn, practice, and review skills all by yourself. Start by opening the Answer Case. Place the number tiles on the top half of the case with the number side up. The letters A–L will be on the bottom. Answer the questions in your activity book, close the case and flip it over, and check to see if your answers match. VersaTiles activities can help you learn about numbers and problem solving as you practice and strengthen your math skills.

Mathematics for SylvanSync™
Grades 3–5
VersaTiles: Add and Subtract With Regrouping

Activity

- Go to page 7 in *VersaTiles: Add and Subtract With Regrouping* to practice finding the sum of two 3-digit numbers with regrouping. Flip over the box to see if you did it correctly.
- Go to page 9 in *VersaTiles: Add and Subtract With Regrouping* to practice finding the sum of three or four 2- to 3-digit numbers with regrouping. Flip over the box to see if you did it correctly.
- Go to page 15 in *VersaTiles: Add and Subtract With Regrouping* to practice finding the difference of two 2-digit numbers with regrouping. Flip over the box to see if you did it correctly.
- Go to page 20 in *VersaTiles: Add and Subtract With Regrouping* to practice finding the difference with regrouping. Flip over the box to see if you did it correctly.

Mathematics for SylvanSync™
Grades 3–5
VersaTiles: Whole Numbers, Decimals, and Fractions

Objectives

The student will be able to:

- write equivalent fractions.
- rename improper fractions as a mixed number.
- write the equivalent decimal for a fraction or mixed number.
- solve a problem that involves naming or comparing decimals.

Materials

- *VersaTiles: Whole Numbers, Decimals, and Fractions* book (VT4WNDF)
- VersaTiles

Background Information

VersaTiles let you learn, practice, and review skills all by yourself. Start by opening the Answer Case. Place the number tiles on the top half of the case with the number side up. The letters A–L will be on the bottom. Answer the questions in your activity book, close the case and flip it over, and check to see if your answers match. VersaTiles activities can help you learn about numbers and problem solving as you practice and strengthen your math skills.

Mathematics for SylvanSync™
Grades 3–5
VersaTiles: Whole Numbers, Decimals, and Fractions

Activity

- Go to page 15 in *VersaTiles: Whole Numbers, Decimals, and Fractions* to practice writing equivalent fractions. Flip over the box to see if you did it correctly.
- Go to page 22 in *VersaTiles: Whole Numbers, Decimals, and Fractions* to practice renaming an improper fraction as a mixed number. Flip over the box to see if you did it correctly
- Go to page 27 in *VersaTiles: Whole Numbers, Decimals, and Fractions* to practice writing the equivalent decimal for a fraction or mixed number. Flip over the box to see if you did it correctly.
- Go to page 30 in *VersaTiles: Whole Numbers, Decimals, and Fractions* to practice solving problems involving naming or comparing decimals. Flip over the box to see if you did it correctly.

Mathematics for SylvanSync™
Grades 3–5
VersaTiles: Patterns and Relationships

Objectives

The student will be able to:

- complete a magic square.
- use function rules and function machines.
- identify an ordered pair or a point in the first quadrant.
- continue a geometric pattern.

Materials

- *VersaTiles: Patterns and Relationships* book (VT4PR)
- VersaTiles

Background Information

VersaTiles let you learn, practice, and review skills all by yourself. Start by opening the Answer Case. Place the number tiles on the top half of the case with the number side up. The letters A–L will be on the bottom. Answer the questions in your activity book, close the case and flip it over, and check to see if your answers match. VersaTiles activities can help you learn about numbers and problem solving as you practice and strengthen your math skills.

Mathematics for SylvanSync™
Grades 3–5
VersaTiles: Patterns and Relationships

Activity

- Go to page 9 in *VersaTiles: Patterns and Relationships* to practice completing magic squares. Flip over the box to see if you did it correctly.
- Go to page 18 in *VersaTiles: Patterns and Relationships* to practice function rules and function machines. Flip over the box to see if you did it correctly.
- Go to page 22 in *VersaTiles: Patterns and Relationships* to practice identifying ordered pairs and points in the first quadrant. Flip over the box to see if you did it correctly.
- Go to page 28 in *VersaTiles: Patterns and Relationships* to practice continuing a geometric pattern. Flip over the box to see if you did it correctly.

Mathematics for SylvanSync™
Grades 3–5
VersaTiles: Probability and Data

Objectives

The student will be able to:

- find the mean, median, or mode of a set of data.
- solve problems using a line graph.
- describe probability of an outcome using a fraction.
- predict outcomes.

Materials

- *VersaTiles: Probability and Data* book (VT4PD)
- VersaTiles

Background Information

VersaTiles let you learn, practice, and review skills all by yourself. Start by opening the Answer Case. Place the number tiles on the top half of the case with the number side up. The letters A–L will be on the bottom. Answer the questions in your activity book, close the case and flip it over, and check to see if your answers match. VersaTiles activities can help you learn about numbers and problem solving as you practice and strengthen your math skills.

Mathematics for SylvanSync™
Grades 3–5
VersaTiles: Probability and Data

Activity

- Go to page 7 in *VersaTiles: Probability and Data* to practice finding the mean, median, or mode of a set of data. Flip over the box to see if you did it correctly.
- Go to page 22 in *VersaTiles: Probability and Data* to practice solving problems using a line graph. Flip over the box to see if you did it correctly.
- Go to page 28 in *VersaTiles: Probability and Data* to practice describing probability of an outcome. Flip over the box to see if you did it correctly.
- Go to page 30 in *VersaTiles: Probability and Data* to practice predicting outcomes. Flip over the box to see if you did it correctly.

Mathematics for SylvanSync™
Grades 3–5
VersaTiles: Expressions and Equations

Objectives

The student will be able to:

- evaluate expressions with variables.
- solve 1-step addition and subtraction equations.
- solve 1-step multiplication and division equations.

Materials

- *VersaTiles: Expressions and Equations* book (VT5EE)
- VersaTiles

Background Information

VersaTiles let you learn, practice, and review skills all by yourself. Start by opening the Answer Case. Place the number tiles on the top half of the case with the number side up. The letters A–L will be on the bottom. Answer the questions in your activity book, close the case and flip it over, and check to see if your answers match. VersaTiles activities can help you learn about numbers and problem solving as you practice and strengthen your math skills.

Mathematics for SylvanSync™
Grades 3–5
VersaTiles: Expressions and Equations

Activity

- Go to page 3 in *VersaTiles: Expressions and Equations* to practice evaluating expressions with variables. Flip over the box to see if you did it correctly.
- Go to page 12 in *VersaTiles: Expressions and Equations* to practice solving 1-step addition equations. Flip over the box to see if you did it correctly.
- Go to page 16 in *VersaTiles: Expressions and Equations* to practice solving 1-step subtraction equations. Flip over the box to see if you did it correctly.
- Go to page 18 in *VersaTiles: Expressions and Equations* to practice solving 1-step multiplication equations. Flip over the box to see if you did it correctly.
- Go to page 21 in *VersaTiles: Expressions and Equations* to practice solving 1-step division equations. Flip over the box to see if you did it correctly.

Mathematics for SylvanSync™

Grades 3–5

The Problem-Solving Process

Objectives

The student will be able to:

- identify the four basic steps of the problem-solving process.
- use the problem-solving process to solve word problems.

Materials

- none

Background Information

Review the four basic problem-solving steps that are used to solve any math problem.

- **Step 1: Understand**
What information is given?
What problem is to be solved?
- **Step 2: Plan**
What is the plan to solve this problem?
- **Step 3: Solve**
Use the plan to solve the problem.
- **Step 4: Look back**
Is the answer reasonable?
Check the answer.

Mathematics for SylvanSync™

Grades 3–5

The Problem-Solving Process

Activity

Review and use the four basic steps to the problem-solving process to solve each problem. Record every step and your answers on a separate sheet of paper.

Part 1

On a separate sheet of paper, fill in the blanks.

1. The four steps of the Problem Solving Process are Understand, _____, _____, and Look Back.
2. In the _____ step, you determine what strategy you will use to solve the problem.
3. Johan asked himself if his answer was reasonable. What step is Johan performing? _____
4. In which step do you use your plan? _____

Part 2

Use the word problem below to answer each question.

Hobo Hut sold 425 pizzas last month. This month it sold 124 fewer pizzas than last month. How many total pizzas were sold in the two months?

1. Which step is shown below?

There were 425 pizzas sold last month.

This month there were 124 fewer sold.

What were the total number of pizzas sold in the two months?

2. Which step is shown below?

Is the answer reasonable?

3. Which step is shown below?

$$425 - 124 = 301$$

$$301 + 425 = 726$$

726 pizzas were sold.

Mathematics for SylvanSync™

Grades 3–5

The Problem-Solving Process

Part 3

Stephanie used the four basic steps to the problem-solving process to solve the following problem. She showed her work on four index cards, but the cards are not in the correct order. Write the name of each step in the correct order and the letter of the correct index card.

There are six seagulls on each balcony on Loren's building. How many seagulls are on 12 balconies?

A. $6 \times 12 = 72$ There are 72 seagulls on the balconies.	B. Does this answer make sense? Divide 72 by 12.
C. There are 6 seagulls on each balcony. There are 12 balconies. How many seagulls are there altogether?	D. Multiply the number of seagulls by the number of balconies.

Mathematics for SylvanSync™
Grades 3–5
The Problem-Solving Process

Part 4

Show your work for each of the four steps for the problem-solving process as you solve each problem.

1. Felipe's company is giving away 32 computers to local schools. Each school will receive 4 computers. How many schools will receive computers?
2. Melinda has \$50.00. She ordered a book for \$22.00 and a DVD for \$12.00. How much money does she have left after her purchases?
3. Bo earns \$8 for each hour he cleans and \$18 for each lawn he mows. This week he cleaned for 4 hours and mowed 3 lawns. How much money did Bo earn this week?
4. Dally's Department Store is trying to sell all of its socks. There are 597 pairs of socks in the storeroom. They are holding a sale that applies if you buy three pairs of socks at a time. Is it possible that all 597 pairs of socks could be sold at the sale price?
5. Apples cost \$1.80 for four. How much would it cost to buy 10 apples?

Mathematics for SylvanSync™

Grades 3–5

Problem-Solving Strategies

Objectives

The student will be able to:

- choose an appropriate problem-solving strategy.
- use problem-solving strategies to solve word problems.

Materials

- none

Background Information

There are many different strategies for solving word problems. Problems can always be solved using more than one strategy. When you know a variety of different strategies, you can check your results by solving the problem in different ways. Some of the strategies are:

Draw a Picture □□□□ + □□□□	Make a Model ↵--120 miles--↵ ↓ 85 miles ↓ ↵	Make a Table or List <table><tr><th>Game</th><th>Points</th></tr><tr><td>2</td><td>8</td></tr><tr><td>4</td><td>12</td></tr><tr><td>6</td><td>16</td></tr></table>	Game	Points	2	8	4	12	6	16
Game	Points									
2	8									
4	12									
6	16									
Find a Pattern ◀▶▶◀◀▶	Guess and Test ?	Work Backwards ↑ ↑ ↑ ↑								

Mathematics for SylvanSync™
Grades 3–5
Problem-Solving Strategies

Read the problem. Choose a strategy and solve.

Jude is researching his family history. He has learned that his great grandfather is 77 years older than he is. His Great Aunt Sasha is 28 years younger than his great grandfather. Jude is 17. How old is his Great Aunt Sasha?

Work backwards to solve this problem.

Jude is 17 years old. His great grandfather is 77 years older than he is. His great grandfather's age can be found by adding 77 to 17. His great grandfather is 94 years old.

Jude's Great Aunt Sasha is 28 years younger than his great grandfather. If his great grandfather is 94 then subtracting 28 from 94 can determine Sasha's age. His great aunt Sasha is 66 years old.

Mathematics for SylvanSync™
Grades 3–5
Problem-Solving Strategies

Activity

Record your answers on a separate piece of paper.

Part 1

Use the table below to solve each problem. Show all work and write your answer in a complete sentence.

Remaining Population of Endangered Species

Endangered Species	Population
Northern Right Whale	350
Red Wolf	100
Siberian Tiger	500
California Condor	130
Florida Panther	80
Mountain Gorillas	700
Amur Leopard	40

1. Which species has the fewest animals remaining?
2. How many fewer red wolves are there than California condors?
3. What is the total population of Amur leopards, Siberian tigers, and Florida panthers?
4. If each of the species doubles its population, what would be the total population of these seven species?
5. Why does this table make solving these problems easier than if they were listed in a paragraph of information?

Mathematics for SylvanSync™
Grades 3–5
Problem-Solving Strategies

Part 2

Select a strategy and solve each problem. Show all work and write your answer in a complete sentence.

1. A marathon committee is setting up water tables along the 42-kilometer marathon route. They will set up 13 water tables at equal intervals. How many kilometers are between each water table?
2. Misty travels 35 miles east of her own house to visit her grandmother. She then drives 45 miles south to visit Aunt Charlene. Misty's next stop is to visit Uncle Rocky who lives 15 miles west of Charlene. How many miles has Misty traveled when she arrives at Uncle Rocky's house?
3. In the first seven games of the season, Bryson scores the following number of points: 4, 8, 7, 11, 10, 14, 13. His sister noticed a pattern in the number of points he has scored. If he continues this pattern, how many points would he expect to score in the tenth game of the season?
4. Marquis has \$16.51 in his bank. He has three bills and nine coins. What are the bills and coins that he has in his bank?
5. Sheila spent 38 minutes running and then spent 49 minutes practicing ballet. She was finished at 5:45 pm. What time did Sheila start running?

Mathematics for SylvanSync™
Grades 3–5
Problem-Solving Strategies

Part 3

Solve each problem using two different strategies. Show all work and write your answer in a complete sentence.

1. Eli and his cousins have agreed to help their grandparents with chores every month. The first month they work for two hours. The second month they work for four hours. The third month they work for six hours. If they continue to follow this pattern, how many hours will they work in the sixth month?
 - Strategy 1:
 - Strategy 2:
2. Natalie scored five points in her first game. In every game after that she scored two points more than she scored in the game before. If this pattern continues, how many points would Natalie score in game six?
 - Strategy 1:
 - Strategy 2:

Mathematics for SylvanSync™

Grades 3–5

Flash Cards

Objectives

The student will be able to:

- demonstrate mastery of basic multiplication facts.
- demonstrate mastery of basic division facts.
- demonstrate mastery of both basic multiplication and division facts.

Materials

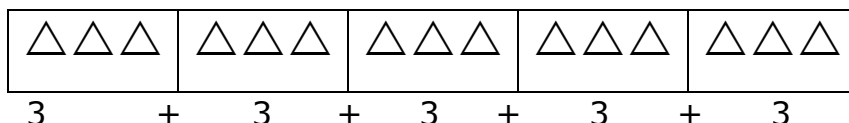
- multiplication and division flash cards
- multiplication and division triangle flash cards
- color tiles, optional

Background Information

Multiplication can be thought of as repeated addition or the sum of equal groups. The answer to a multiplication problem is the **product**.

Gail gave three biscuits to each of her five dogs. How many treats did Gail give to her dogs?

Gail gave five groups of three biscuits to her dogs.

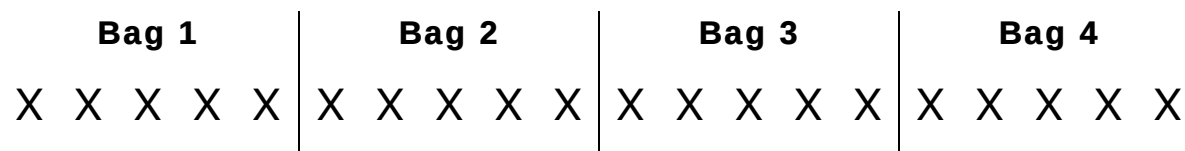


$$5 \times 3 = 15$$

Gail gave 15 biscuits to her dogs.

Division can be thought of as separating one group into equal groups. The answer to a division problem is the **quotient**.

Jason made 20 sandwiches and divided them equally into four bags. How many sandwiches are in each bag?



$$20 \div 4 = 5$$

There are 5 sandwiches in each bag.

Mathematics for SylvanSync™

Grades 3–5

Flash Cards

Activity

Use the flash cards to multiply and divide. If desired, model the number sentences using color tiles. Record your answers on a separate sheet of paper.

Part 1

Shuffle the deck of the multiplication flash cards. Pick a card. Find the product. Check the answer on the other side of the card.

Write the number sentence and the answer. Repeat until you have answered 20 cards correctly.

Part 2

Shuffle the deck of the division flash cards. Pick a card. Find the quotient. Check the answer on the other side of the card.

Write the number sentence and the answer. Repeat until you have answered 20 cards correctly.

Part 3

Shuffle the deck of triangle flash cards. Arrange the cards so the purple numbers are at the top. Draw a card. Use a small piece of paper or your hand to cover the purple number.

Write two multiplication number sentences using the two pink numbers. Find the products. Uncover the purple number to check your answers. Repeat until you have answered 20 cards correctly.

Part 4

Shuffle the deck of triangle flash cards. Draw a card. Use a small piece of paper or your hand to cover up one of the pink numbers.

Write the division number sentence using the purple and pink numbers. Find the quotient. Uncover the third number to check your answer. Repeat until you have answered 20 cards correctly.

Mathematics for SylvanSync™

Grades 3–5

Flash Cards

Part 5

Draw five triangles on your paper. Place a division sign in the middle of two of the sides and a multiplication sign in the middle of the third side.

Shuffle the deck of triangle flash cards. Draw a card. Look at the three numbers on the card. Insert the numbers in the correct corners in the triangle on your paper.

Repeat until your triangles are completed.

Part 6: *Optional*

Shuffle the triangle flash cards. Draw a card. On your paper write the multiplication and division facts related to the three numbers.

Repeat 10 times with 10 different cards.

Mathematics for SylvanSync™

Grades 3–5

Pascal's Triangle

Objectives

The student will be able to:

- identify multiples of three.
- identify multiples of four.

Materials

- colored pencils
- one copy of Worksheet 1
- two copies of Worksheet 2

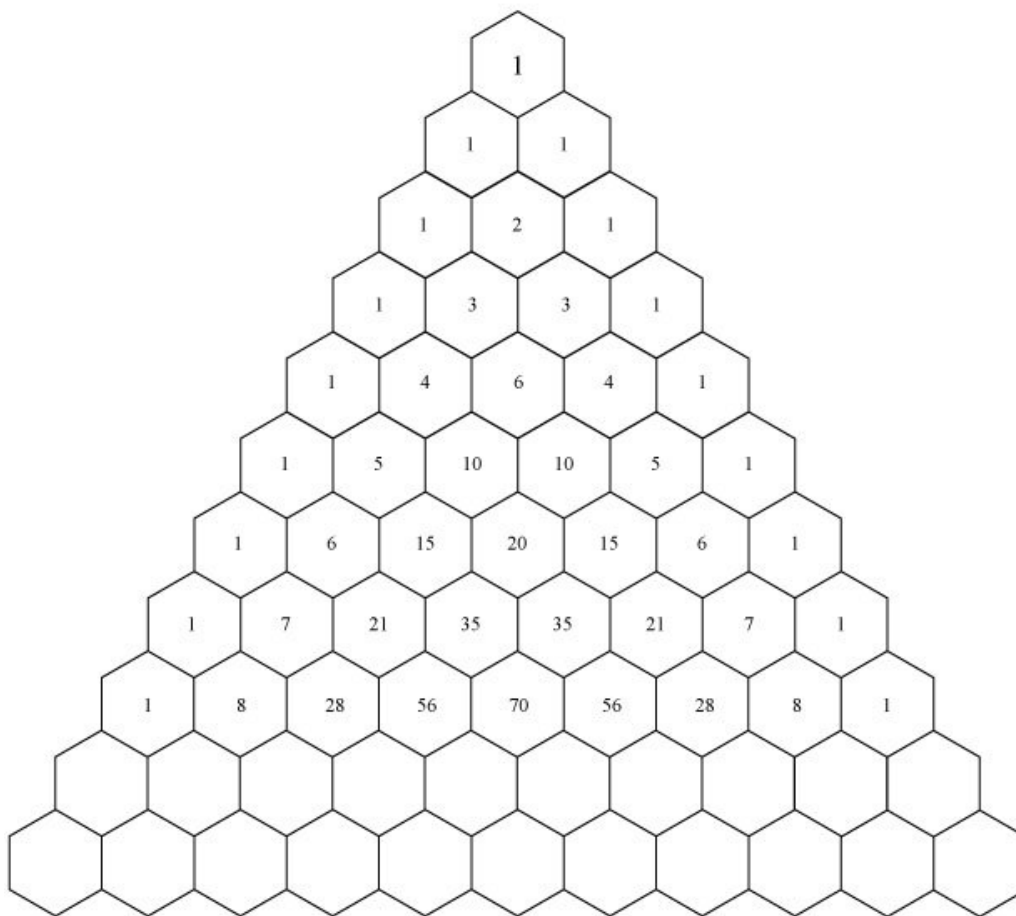
Background Information

Pascal's triangle is an arrangement of numbers that reveal many important and interesting patterns. Look at Pascal's Triangle on the next page.

Mathematics for SylvanSync™

Grades 3–5

Pascal's Triangle



Are there patterns within each row?

- The numbers repeat from the center of each row out towards the left and the right.

Is there a pattern in the diagonal numbers?

- The counting numbers can be found in two diagonals.

Many more patterns can be found in Pascal's Triangle.

Mathematics for SylvanSync™

Grades 3–5

Pascal's Triangle

Activity

Use Worksheets 1 and 2 to explore patterns and multiples in Pascal's Triangle. Record your answers on a separate sheet of paper.

Part 1

Use Worksheet 1 to complete the following exercises.

1. Place two fingers on two numbers that are side-by-side. What is true about the number below those numbers? Repeat two more times to test your theory. Explain.
2. How can one row be used to find the numbers for the next row? Explain.
3. Complete the empty rows of the triangle.

Part 2

Use two copies of Worksheet 2 to complete the exercises.

1. Use a copy of Worksheet 2 and color all the multiples of 3 in the triangle.
2. Use another copy of Worksheet 2 and a second color to color all the multiples of 4 in the triangle.

Part 3

Place the triangle with the multiples of 4 colored on top of the triangle with the multiples of 3 colored. Hold them up to a light.

1. What is true about the spaces with the number 36 in them? What does this mean?
2. Are there other numbers that overlap in coloring?
3. Name other numbers that are multiples of 3 and 4.

Part 4

Color the white spaces in the two triangles all one color.

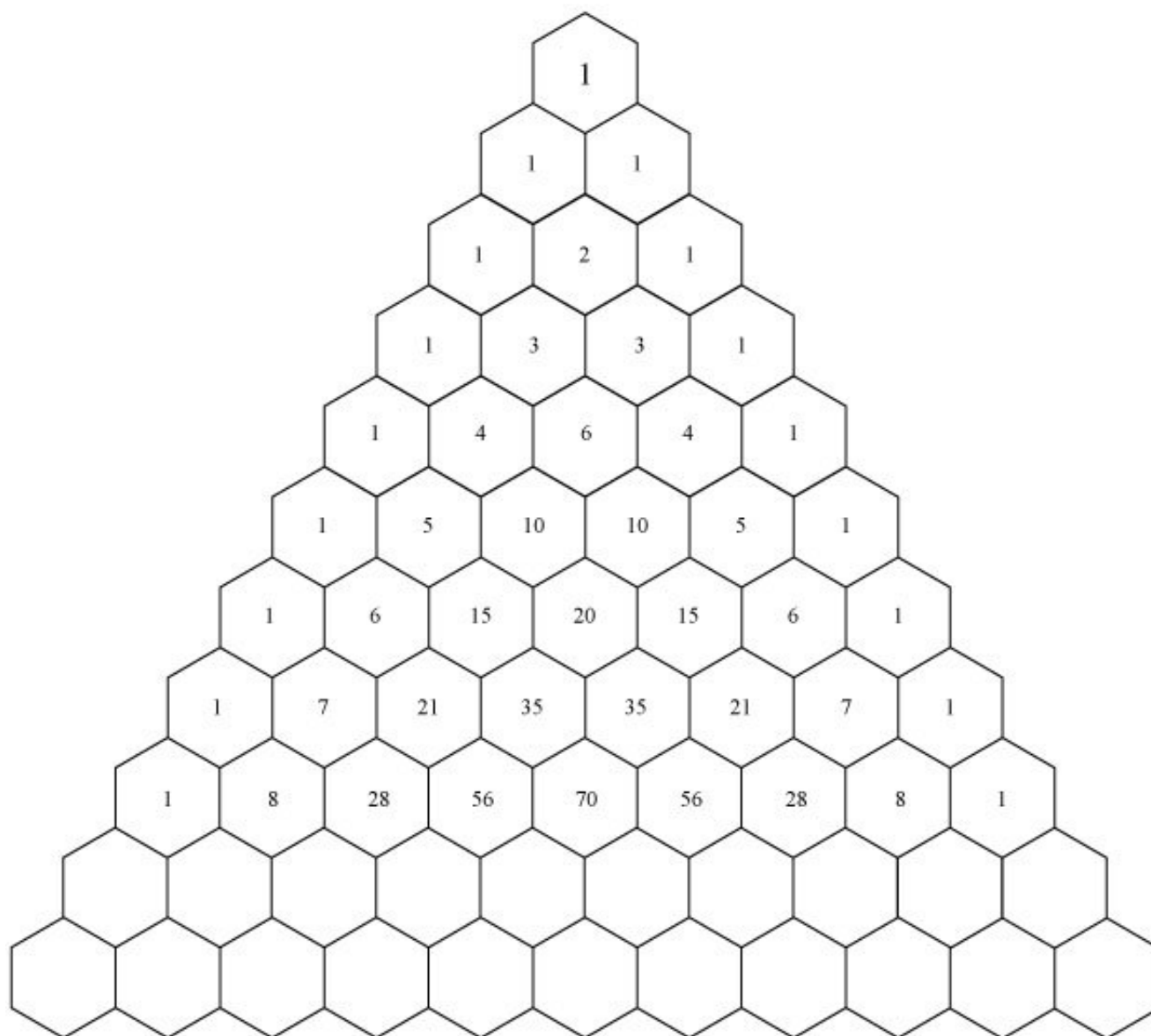
1. Do you see a pattern between the two worksheets?
2. Can you predict what the triangle with the multiples of 5 would look like? Draw a sketch.

Part 5: *Optional*

Ask for an additional copy of Worksheet 2. Color the multiples of 5 and check your prediction.

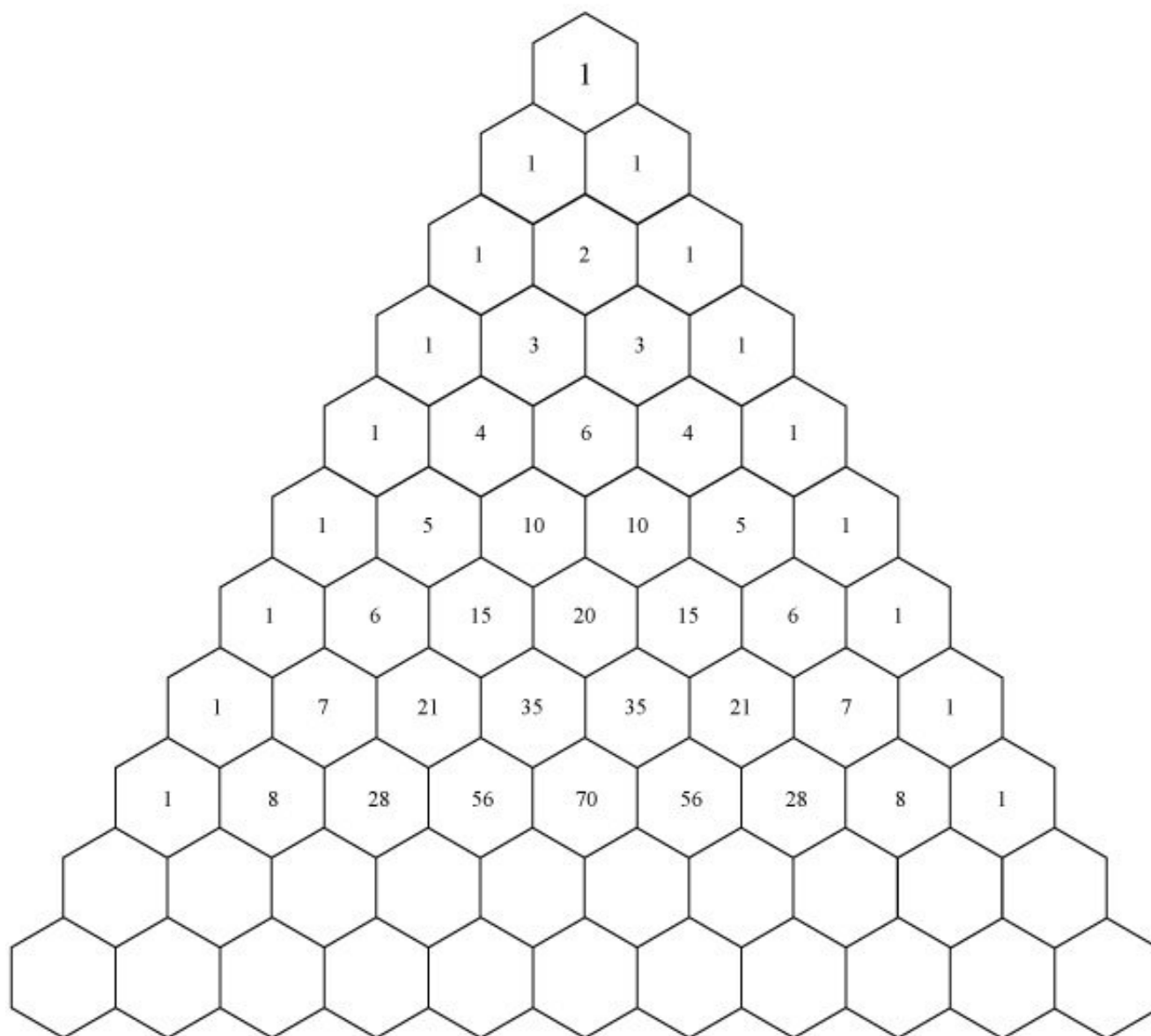
Mathematics for SylvanSync™
Grades 3–5
Pascal's Triangle

Pascal's Triangle Worksheet 1



Mathematics for SylvanSync™
Grades 3–5
Pascal's Triangle

Pascal's Triangle Worksheet 2



Mathematics for SylvanSync™

Grades 3–5

Understanding Decimals

Objectives

The student will be able to:

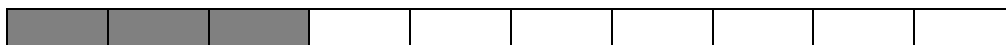
- model and name a decimal to the hundredths.
- write the equivalent decimal for a fraction.
- write the equivalent fraction for a decimal.

Materials

- Decimal Tower Cubes
- Fraction Tower Cubes
- colored pencils
- graph paper

Background Information

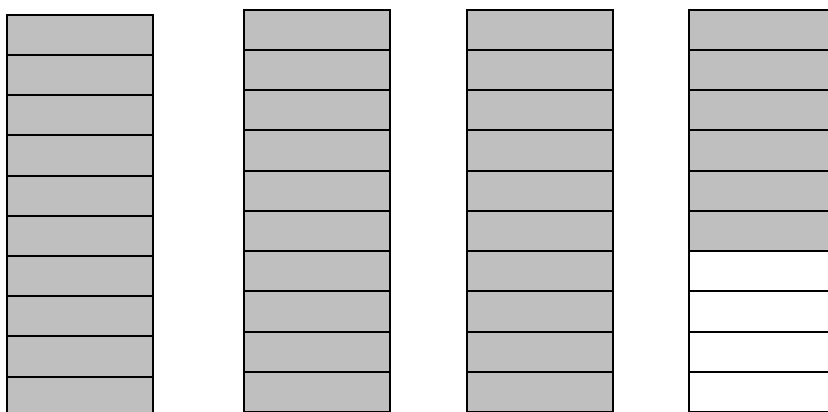
A decimal represents a fractional part of a number.



Three out of the 10 parts are gray. Three-tenths of the model are shaded.

As a decimal it is written as 0.3. As a fraction it is written as $\frac{3}{10}$.

Hal used 3 sheets of paper plus six-tenths of another sheet for his report. This is written as 3.6.



Mathematics for SylvanSync™
Grades 3–5
Understanding Decimals

Activity

Record your answers on a separate sheet of paper.

Part 1

Draw a model of each of the following.

1. Seven-tenths of the buttons were red.
2. The children ate 2.4 candy bars. Hint: Think 2 whole and four-tenths.
3. Julie painted nine-tenths of the wall purple and one-tenth green.
4. There were 3.5 pizzas left when the team was finished eating.

Part 2

Follow the directions to draw a shaded grid on your graph paper. Write a decimal for the shaded region.

1. Draw a square that is 10 units by 10 units and shade 56 units.
2. Draw three squares that are 10 units by 10 units. Completely shade two of the squares and 11 units in the third square.
3. Draw a square that is 10 units by 10 units and shade 6 units.
4. Draw a square that is 10 units by 10 units and shade 99 units.
5. Draw two squares that are 10 units by 10 units. Completely shade one of the squares and 78 units in the other square.

Mathematics for SylvanSync™
Grades 3–5
Understanding Decimals

Part 3

Use Decimal Tower Cubes and Fraction Tower Cubes to complete the exercises.

1. Find the fraction equivalent to the given decimal.

- a. 0.2
- b. $0.3\overline{3}$
- c. 0.125
- d. 0.75
- e. 0.6
- f. 0.875
- g. $0.16\overline{6}$

2. Find the decimal equivalent to the given fraction.

- a. $\frac{1}{4}$
- b. $\frac{1}{12}$
- c. $\frac{3}{6}$
- d. $\frac{7}{10}$
- e. $\frac{2}{3}$
- f. $\frac{5}{8}$

Mathematics for SylvanSync™
Grades 3–5
Understanding Decimals

Part 4

Use Decimal Tower Cubes and Fraction Tower Cubes to complete the exercises.

Copy and complete the chart.

	Answer Given as a Fraction	Answer Given as a Decimal
1. $\frac{1}{10} + 0.2 =$		
2. $0.5 + \frac{1}{4} =$		
3. $0.08\bar{3} + \frac{11}{12} =$		
4. $\frac{1}{8} + 0.25 =$		

Mathematics for SylvanSync™

Grades 3–5

Understanding Fractions

Objectives

The student will be able to:

- write equivalent fractions.
- write fractions in simplest form.
- add and subtract fractions with like denominators.

Materials

- Fraction Tower Cubes
- Fraction Circles
- Fraction Tiles

Background Information

Ashanti has a whole pie which she cuts into six equal pieces. Soon most of the pie is gone. The shaded section shows what portion of the pie is left.



There is $\frac{1}{6}$ of the pie left, while $\frac{5}{6}$ of the pie has been eaten.

Her aunt says if she had cut the pie into 12 pieces, the portion remaining would be the same. Is this true?

There is $\frac{2}{12}$ of the pie left, while $\frac{10}{12}$ of the pie has been eaten.

$\frac{2}{12}$ is equivalent to $\frac{1}{6}$

Mathematics for SylvanSync™
Grades 3–5
Understanding Fractions

Activity

Record your answers on a separate sheet of paper.

Part 1

Use the Fraction Circles to complete the exercises.

1. Place the red whole circle on your paper. Find four ways to equal one whole with the Fraction Circles. Sketch and color each answer according to the colors of the parts. Write an addition number sentence for each sketch.
2. Use Fraction Circles to find three fractions equivalent to $\frac{2}{3}$.
3. Find the Fraction Circle sections that represent $\frac{1}{2}$ and $\frac{1}{4}$. Tanja says that in order to add these together they should be in equal parts. Explain and find the sum.

Mathematics for SylvanSync™

Grades 3–5

Understanding Fractions

Part 2

Use Fraction Tiles to complete the exercises.

1. Set aside the red tile that equals a whole. Turn all other tiles face down and mix them up. Pick a tile. Explain how many more are needed to make a whole.
2. Find six tiles marked $\frac{1}{8}$. What does that equal? How can this fraction be represented using the fewest tiles? This is called the simplest _____ of a fraction.
3. Model $\frac{10}{12}$ with the black tiles. Represent that fraction in its simplest form.
4. Fran says that the following fractions cannot be added together. Use your tiles to prove her wrong. Explain and find the sum. Write the answer in simplest form.

$$\frac{1}{6} + \frac{1}{10} + \frac{1}{10} + \frac{1}{5} + \frac{1}{10}$$

Mathematics for SylvanSync™
Grades 3–5
Understanding Fractions

Part 3

1. Use Fraction Cubes to determine the missing parts of the equivalent fractions.

a. $\frac{1}{6} = \frac{?}{12}$

b. $\frac{1}{4} = \frac{?}{12} = \frac{?}{8}$

c. $\frac{6}{10} = \frac{?}{5}$

d. $\frac{8}{12} = \frac{?}{6} = \frac{?}{3}$

2. Using the Fraction Cubes, rewrite the pairs of fractions so they have like denominators. Then find the sum.

a. $\frac{1}{4}$ and $\frac{3}{8}$

b. $\frac{2}{3}$ and $\frac{1}{6}$

c. $\frac{2}{5}$ and $\frac{3}{10}$

Mathematics for SylvanSync™
Grades 3–5
Understanding Fractions

Part 4

Use Fraction Cubes, Circles, or Tiles to find the sum or difference. Write each answer in its simplest form.

1. $\frac{2}{6} + \frac{1}{3}$

2. $\frac{3}{4} - \frac{7}{12}$

3. $\frac{1}{2} + \frac{1}{3} + \frac{2}{12}$

4. $1 - \frac{3}{5}$

5. $\frac{11}{12} - \frac{4}{8}$

6. $\frac{1}{2} + \frac{1}{6} - \frac{5}{12}$

Mathematics for SylvanSync™

Grades 3–5

Plane Shapes

Objectives

The student will be able to:

- name polygons and represent them on a Geoboard.
- determine the perimeter of a polygon represented on a Geoboard.
- determine the area of a polygon represented on a Geoboard.

Materials

- Geoboard
- rubber bands

Background Information

The **perimeter** of a polygon is the distance around the polygon. The **area** of a polygon is the number of square units that can fit inside the polygon.

Use a rubber band to make a rectangle on your Geoboard that has three units on one side and four units on the other side.

The distance around the rectangle is 3 units + 4 units + 3 units + 4 units.
The perimeter is 14 units.

Count the square units inside the rectangle in order to determine the area.
The area is 12 square units.

Mathematics for SylvanSync™
Grades 3–5
Plane Shapes

Activity

Record your answers on a separate sheet of paper.

Part 1

Represent each of the following on a Geoboard. Create a sketch your work before you move on to the next question.

1. two squares
2. a square and a triangle
3. a rectangle and a rhombus
4. a trapezoid and a triangle
5. a hexagon and a square

Part 2

Represent each of the following on a Geoboard. Sketch the polygon on your paper and answer each question.

1. a triangle with a height of 4 units and base length of 4 units
What is the area of this triangle?
2. a rectangle with a width of 2 units and length of 7 units
What are the perimeter and area of this rectangle?
3. a trapezoid with bases of 4 units and 6 units, and a height of 3 units
What is the area of this trapezoid?
4. a parallelogram with two sides of 7 units and a height of 4 units
What is the area of this parallelogram?

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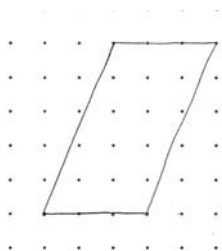
Grades 3–5

Plane Shapes

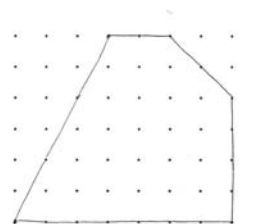
Part 3

Model each polygon on your Geoboard, and find the area of each.

1.



2.



Part 4

Use your Geoboard to create polygons that meet the following requirements. Sketch your results.

1. a square with a perimeter of 4 units
2. a rectangle with a perimeter of 18 units
3. a triangle with an area of 9 square units
4. a hexagon with an area of 42 square units
5. a trapezoid with an area of 24 square units
6. a rectangle with a perimeter of 26 units and an area of 36 square units
7. Create the largest hexagon possible on the board. Find the area.
8. Create the largest triangle possible on the board. Find the area.
9. Create the largest rectangle possible on the board. Find the area and the perimeter.

Mathematics for SylvanSync™

Grades 3–5

Shapes in Space

Objectives

The student will be able to:

- identify geometric shapes.
- identify vertices, edges, and faces.

Materials

- Geometric Solids

Background Information

Study the solids in your set of Geometric Solids. Spend some time holding them, rolling them, stacking them, and identifying them before you get started.

Can you identify the different solids?

cube

cone

cylinders

sphere

hemisphere

square pyramid

hexagonal prism

octagonal prism

triangular prism

rectangular prisms

Remember that solids have vertices, edges, and faces. **Faces** are the flat sides. **Edges** join the faces. And the **vertices** are the corners, where the edges join.

Mathematics for SylvanSync™
Grades 3–5
Shapes in Space

Activity

Use the Geometric Solids to complete the exercises. Record your answers on a separate sheet of paper.

Part 1

Who Am I? Name the solid described.

1. I have no faces and I can roll!
2. I can roll but I have two faces.
3. I am one of two shapes here that has just one flat side. I am half of one of the other solids.
4. I have 8 faces.
5. I have 5 faces, 6 vertices, and 9 edges.
6. I have 16 vertices and 24 edges!
7. I have 5 faces, 5 vertices, and 8 edges.

Part 2

Describe the similarities and differences between the pairs of solids.

1. the two rectangular prisms
2. the two cylinders
3. the sphere and the hemisphere
4. the cube and the rectangular prism
5. the triangular prism and the pyramid
6. the cone and a cylinder

Mathematics for SylvanSync™

Grades 3–5

Shapes in Space

Part 3

Copy the table onto your paper. Put all the solids in a container. With your eyes closed, reach in and pick a solid. Keeping your eyes closed, examine the solid with your hands. Feel the number of faces, edges, and vertices. See if there are other features of the solid that help you identify it. Put the solid in your lap or out of sight while you fill in the table. Do not replace the solid. Continue until all the solids have been selected. Check your work by studying each solid.

	Number of Faces	Number of Edges	Number of Vertices	Other Features	Name of Solid
1.					
2.					
3.					
4.					
5.					
6.					
7.					
8.					
9.					
10.					
11.					
12.					

Mathematics for SylvanSync™

Grades 3–5

Probability Activities

Objectives

The student will be able to:

- identify all possible outcomes of an event.
- determine the experimental probability of an event.

Materials

- spinners
- number cubes

Background Information

Probability is the chance that an event will happen. The ratio used to determine probability is:

$$\frac{\text{number of ways an event can happen}}{\text{total number of possible outcomes}}$$

When you roll a number cube, what is the chance that it will land on the number 5? What are the possible outcomes when you roll the number cube? The total possible outcomes when you roll a number cube are the numbers 1, 2, 3, 4, 5, or 6.

$$\frac{\text{number of ways an event can happen}}{\text{total number of possible outcomes}} = \frac{1 \text{ way to roll a five}}{6 \text{ possible outcomes}} = \frac{1}{6}$$

The probability of rolling the number 5 is $\frac{1}{6}$.

Experimental probability is the chance that an event will happen based on an experiment and those results.

Nichole rolled a number cube five times. She rolled the numbers 4, 5, 3, 4, and 2. What is the experimental probability of rolling the number 4?

The ratio for experimental probability is:

$$\frac{\text{number of times the desired outcome occurred}}{\text{total number of outcomes}} = \frac{\text{rolled a four 2 times}}{\text{rolled the die 5 times}} = \frac{2}{5}$$

Mathematics for SylvanSync™
Grades 3–5
Probability Activities

Activity

Complete each exercise. Record your answers on a separate sheet of paper.

Part 1

List the number of possible outcomes for each situation.

1. landing on pink with a spinner that has equal sections colored yellow, orange, purple, and pink
2. selecting a queen of hearts from a deck of 52 cards
3. flipping a coin
4. picking a number from 1 to 10

Part 2

Determine the probability for the following situations.

1. A bag contains 5 red marbles, 7 black marbles, 8 green marbles, and 10 yellow marbles. What is the probability of:
 - a. picking a red marble?
 - b. picking a black or green marble?
 - c. picking a white marble?
2. Each of the 26 letters in the alphabet are written on separate cards and then placed in a hat. What is the probability of:
 - a. picking one of the main vowels (a, e, i, o, u)?
 - b. picking a consonant?
 - c. picking the letter N or M?

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Grades 3–5

Probability Activities

Part 3

Use a number cube to perform the experiments to determine experimental probability. Copy each chart and record your answers on your paper.

Before you roll the number cube record the probability that each number will be rolled. Then roll the number cube 20 times, and use tally marks to record the results. Determine the experimental probability.

Number	1	2	3	4	5	6
Probability (one roll)	$\frac{1}{6}$					
Record 20 rolls						
Experimental probability						

Before you roll the number cube, record the probability that an even or odd number will be rolled. Then roll the number cube 20 times, and use tally marks to record the results. Determine the experimental probability.

Number	Odd	Even
Probability (one roll)		
Record 20 rolls		
Experimental probability		

Mathematics for SylvanSync™

Grades 3–5

Probability Activities

Part 4

Use spinners to perform experiments to determine experimental probability. Copy each chart and record your answers on a separate piece of paper.

Before spinning, record the probability that the spinner will land on each number. Then spin the spinner 20 times, and use tally marks to record the results. Determine the experimental probability.

Spinner with Numbers 0–9

Number	0	1	2	3	4	5	6	7	8	9
Probability (one spin)										
Record 20 spins										
Experimental probability										

Before spinning, record the probability that the spinner will land on each color. Then spin the spinner 20 times, and use tally marks to record the results. Determine the experimental probability.

Spinner with Four Colors

Color	Red	Yellow	Green	Blue
Probability (one spin)				
Record 20 spins				
Experimental probability				

Mathematics for SylvanSync™
Grades 3–5
Probability Activities

Spinner with Six Colors

Color	Green	Orange	Purple	Yellow	Red	Blue
Probability (one spin)						
Record 20 spins						
Experimental probability						

Part 5: Optional

Look closely at this spinner. How does this spinner differ from the others? Does this design change the probability? Explain.

Before spinning, record the probability that the spinner will land on each color. Then spin the spinner twenty times and use tally marks to record the results. Determine the experimental probability.

Spinner with Three Colors

Color	Red	Green	Blue
Probability (one spin)			
Record 20 spins			
Experimental probability			

Mathematics for SylvanSync™

Grades 3–5

Wrap-ups: Concept Practice

Objectives

The student will be able to:

- demonstrate mastery of basic multiplication and division facts.
- identify fractional parts and equivalent fractions.
- add and subtract fractions with like denominators.
- identify decimal equivalents of fractions.

Materials

- Learning Wrap-Ups, Multiplication
- Learning Wrap-Ups, Division
- Learning Wrap-Ups, Fractions

Background Information

Wrap-Ups let you practice and review math facts all by yourself. Start by selecting the math facts that you would like to practice. Hold the specific board you want to use at the top. Place the string at the start arrow and answer the first problem. Wrap the string from the problem to the answer and around the back. Continue with each problem, wrapping the string from each problem to the correct answer. When the board is complete, flip it over and check to see if your string matches the marks on the back.

Mathematics for SylvanSync™
Grades 3–5
Wrap-ups: Concept Practice

Activity

- Choose Wrap-Ups Multiplication to practice multiplication facts. Flip over the board to see if you answered the problems correctly. Continue through the multiplication facts.
- Choose Wrap-Ups Division to practice division facts. Flip over the board to see if you answered the problems correctly. Continue through the division facts.
- Choose Wrap-Ups Fractions to practice fraction concepts. Flip over the board to see if you answered the problems correctly. Continue through the concepts and skills on the fraction boards.